

LIQUID eKYC Applicant SDK Specifications

Version 1.44.0

Liquid, INC.

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Revision History

Version	Overview	Details
1.44.0	Added sequence diagrams	Added the following sequence diagrams • [EN]Overall_Comply to Wa_ (Evaluation version) v20260109 • [EN] Comply to "To"/"Chi" (IC Scan through SDK(Evaluation version)) v20260109
1.44.0	Corrected errors in the overall overview diagram	Comply to "Ho" / Comply to "He" selection → eKYC verification method selection
1.44.0	Added the OCR-based input assistance function for authentication information	Available when reading a National Identity Card through the Document IC Card Function
1.44.0	Added Face Verification	Incorporated the content of "【For designated businesses】 [EN]LIQUID eKYC Applicant SDKver Specification (Face Verification ver) v.1.41.0" into this document
1.42.0	Detailing the functions of Apple Wallet	Added an explanation of Apple Wallet
1.42.0	Android 16KB page size support	Added a note regarding the supported environment for Android
1.42.0	Modification of sequence diagrams	Revised the sequence diagram below Corrected the transition destination when a PIN error occurs during IC reading • [EN]Overall_Comply toWa_v20251203.pdf • [EN] Overall_JPKI+(Face)_20251203.pdf • [EN]Overall_Comply to He_v20251203.pdf
1.41.0	Specified the Swift version	Specified that the items below are supported as of this writing • Swift5
1.41.0	Error corrections	Non-forwardable mail (corrected an error in the Japanese text only)
1.40.0	Added functionality to not return the My Number	When the ID document type is a National Identity Card (+ My Number), add instructions for controlling the return of the My Number from the function or Connector.
1.38.0	Newly added function to obtain My Numbers in the "Comply to 'He' method (no screen)" configuration	In the "Comply to 'He' method (no screen)" configuration, the My Number is obtained and returned when using the National Identity Card via the Comply to "He" method.
1.37.0	Supports license information embedded in the National Identity Card	Enabled additional reading of license information embedded in the National Identity Card (commonly referred to as the "MyNa Driver's License") for the following functions: • Document IC Card Function • Document IC card function (without screen) • JPKI Function • JPKI Function (without screen)

1.37.0	Explanation provided for when the Change Display Language function setting does not work	Described actions taken by Google Play and countermeasures implemented within the client's app
1.37.0	Modification of sequence diagrams	Corrections were made to the following sequence diagram: Moved the position of the Activate Function above the completion screen • [JP] Overall, JPKI+(Face) v20250514.pdf • [JP] Overall, comply to "He" v20250514.pdf
1.37.0	Added AppleWallet functionality	The following was added to the "Scope of Functions to be Provided": • Comply to "Ru" (card-substitute electromagnetic record) • Comply to "Ru" + Face
1.33.0	Added precautions on the Comply to "He" method *without screen	Added note that the function is not compatible with National Identity Cards (+ My Number), Residence Cards, and Special Permanent Resident Certificates
1.33.0	Sequence diagram for functions without screen	The following sequence diagram was considered a sales enquiry • Comply to "He" *without screen • JPKI + (MyNumber) *without screen
1.33.0	Added supported languages for the Change Display Language function	Eight languages were added as available languages.
1.33.0	Added description of smartphone JPKI	A reference to smartphone JPKI was added in Section 9: 'Interface'
1.31.0	Precautions added on Comply to "He" method (without screen)	Precautions listed
1.31.0	Modification of sequence diagrams	Corrected a misprint in the description below SE21205 in 'Overall "Wa" method'.
1.30.0	Expanded description of functions without screens	Expanded descriptions of the following functions • Document IC card function (without screen) "verifyIdChipNonInteractive()" • JPKI function (without screen) "identifyIdChipNonInteractive()"
1.30.0	Added sequence diagram	The following sequence diagrams were added • Without screen、JPKI+ (MyNumber)
1.30.0	Moved sequence diagram	Sequence diagrams moved from the Appendix to Chapter 6
1.30.0	Added description of the "Display language change function"	Mentioned that it is not possible to change the language of some statements, such as statements under OS control
1.30.0	Misprint corrections	National Identity Card (+ Get MyNumber) → National Identity Card (+ MyNumber)
1.30.0	Misprint corrections	Deleted description of Residence Card/Special Permanent Resident Certificate from Document IC card function (without screen) "verifyIdChipNonInteractive()"
1.29.0	Display language change function available	Added display language change function *Use is optional
1.28.0	In the document IC card function, the eKYC method called Non-legal IC scan system (without digital signature verification and	Regarding the IC card function, a feature has been implemented that allows it to be used even without a communication environment during the execution of the function. *Available when digital signature verification is not required

	without face photo, offline) is added	
1.28.0	Additional explanation of the confirmation screen	Newly added description of the screen for checking the photos taken for the document IC card function
1.28.0	Modification of sequence diagrams	Misprint correction: Reversed the order of the expiration date check and the electronic signature verification check (to match the actual logic). *Changed from [EN]Overall_Comply to He_ 20240408.pdf to [EN]Overall_Comply to He_ 20240822.pdf
1.28.0	Modification of sequence diagrams	Misprint correction : Changed from 禁輸 accounts to 金融 accounts *Changed from [EN] Comply to ToChi(Photoshoot through SDK) v20240408.pdf to [EN] Comply to ToChi(Photoshoot through SDK) v20240822.pdf
1.27.0	Addition of the comply to He (My Number)	In addition to the comply to He, a My Number is obtained and returned at the same time
1.26.0	Added Document IC card function (without screen)	Existing functions are not affected.
1.25.0	Added the description of JPKI (smartphone JPKI) function	Specified that only Android is supported.
1.25.0	Added the JPKI function (without screen)	Existing functions are not affected.
1.23.0B	Added “JPKI+ (My number)” , “ JPKI+ (Face, My number)”	Obtain and return your personal number (My number) at the same time as JPKI.
1.23.0A	Update Sequence Diagrams (Extended validity time of authentication token)	60min.→90min.
1.23.0	Extended validity time of authentication tokens	60min.→90min.
1.23.0	Error corrections	○○-support → ○○method (notation in Japanese changes)
1.23.0	Sequence diagrams are now in a separate file.	Sequence diagrams at the end of the book are referenced to a separate file, and the diagrams themselves are no longer included.
1.21.0	Added Special Permanent Resident Certificate	Added the following as supported documents • Special Permanent Resident Certificate
1.21.0	More details have been included within the specifications of the Status Auto-verification process.	The specifications of Status Auto-Verification Process for the comply to "He" and "To/Chi" (IC Scan through SDK) of the resident card have been detailed.
1.20.0	Added JPKI (+Face)	JPKI function which includes facial recognition is added.
1.19.0	Added Supported documents	The following documents have been added as supported documents • Student ID Card • Employee ID card • Student handbook • Pension Book • Health insurance card • Basic Resident Registration Card

		• Basic Pension Number Notification • Special Education Book • Passport (Any issuance date) • Special Education Card • Physical Disability Certificate Book • Physical Disability Certificate Card • Mental Disability Certificate Book • Mental Disability Certificate Card
1.19.0	Added Supported methods	The following methods have been added • Comply to front and back(face) • Comply to front and back(no-face))
1.19.0	Corrected an error in the sequence diagram	Corrected an omission of the character string "KYC operations using your system" in the sequence diagram of the Comply to "He".
1.18.0	Added smart phone JPKI	Added new functions," Public Certification (smart phone JPKI) function"
1.17.0	Passive Detection added to the Facial Recognition Function	Added a new shooting method that does not require facial muscle shooting and flash shooting (called "Passive Facial Liveness Detection") *Accordingly, the existing shooting method that requires facial muscle and flash shooting is referred to as active detection.
1.16.0	Added design setting function	Added a new function that allows setting of some colors in the screen.
1.15.3	Added confirmation screen	For Document Function and Facial Recognition Function, a new screen has been added for checking photos.
1.15.1	Change of terms	Before revision : Master setting "New name or address on driver's licenses". "ON" "OFF" After revision : Setting "Whether or not the changed name/address is obtained on the driver's license." "Retrieve the information after the change". Change to "Do not retrieve changed information".
1.15.0	Added description of document IC card function	Specify the behavior when the pre-set is "Whether or not the changed name/address is obtained on the driver's license".
1.14.2	Get Version function released.	Get version function included. *Optional
1.14.1	Screen image removed.	Updated to be consistent with the product manual.
1.14.1	Sequence diagrams organized.	A recommended sequence diagram included for compliance with the Act on Prevention of Transfer of Criminal Proceeds (noting the use of the Comply to "Wa/He/Ho").*Sequence diagrams for use in cases other than the above descriptions follow the previous version.
1.14.0	"Face" photoshoot method in the Document Function released.	Document Function argument updated so that the "Face" photoshoot method is available, in addition to the Comply to "Ho", etc.

1.13.0	Comply to "To/Chi" (IC scan through SDK) in the Document Function released.	Document Function argument updated so that Comply to "Ho", Comply to "To" and Comply to "Chi" are brought in when applicable.
1.12.0a	All screens updated in the Terms of Use.	All screens updated in the Terms of Use with the latest images.
1.12.2	All screens updated for Document IC card Function.	All screens updated for Document IC card Function with the latest images.
	All screens updated for JPKI (Public Certification).	All screens updated for JPKI (Public Certification) with the latest images.
1.12.0	"Terms of use" screen released	Added Terms of Use Function
	"Comply to 'To'/'Chi'" features released	"Comply to 'He'" and "Comply to 'To'/'Chi'" can now be specified as arguments in the Document IC Card Function
	Added descriptions for Initialization Function (Trial Version)	Specified how the function should be used
1.11.0a	Fix sequence diagram.	Fix sequence diagram.
1.11.0	Upgraded security level of "Comply to 'He'"	Upgraded the security level of "Comply to 'He'" without changing its interface or functionalities
	Simplified version of "Comply to 'He'" released (no signature verification)	Added function that does only chip scans for Driver's License, Individual Number Card and Residence Card * This not officially "Comply to 'He'" because it does not include signature verification
	Signature verification-only version of "Comply to 'He'" released (nothing on screen)	Added function that does only signature verification after the chip has been read
	Added specifications for integration in 8.	Clarified specifications such as what function should be used in "Comply to 'Ho'"
	Transferred the sequence diagrams to a separate document	Created a separate document for the sequence diagrams
	Clarified the data storage policy for "Comply to 'Wa'"	Clarified the data storage policy for both Liquid and our clients
1.10.0	"Comply to 'Wa'" features released / Public Certification Service for Individuals is now available	Added function for the Public Certification Service for Individuals (JPKI Function)
1.9.0	Fix the sequence diagram for Comply to "He".	- Switched the order of the activation function section and the application information transmission section. - The OCR section is now within the 90-minute time limit for authentication tokens.

		- Specified that the OCR section is only for Residence Cards.
1.70	[EN] Initial Version	
1.60	Updated 4. Operating Environment	Updated the section to match “LIQUID eKYC Service Specifications”.
	Transferred 9. Interface to a separate document	Created a separate, more detailed document on the interface, formerly included in this document as a table.
	Restrictions to “Comply to ‘He’” for iPhone environments	Added section on Driver’s License restrictions on iPhone7/7Plus.
	Added section on external characters for “Comply to ‘He’”	Clarified external character handling on documents.
	Trial version released	Added sections for the trial version.
	Header font color settings for “Comply to ‘He’”	Added setting instructions for the “Comply to ‘He’” screens.
1.55	“Comply to ‘He’” features released	Added sections for “Comply to ‘He’” * There is no need for modifications unless you plan to use the “Comply to ‘He’” features. • Added a section for the Comply to “He” document ID card function. This function is analogous to the document function of Comply to “Ho”. Additionally, we modified the existing sections on the document function (such as adding clarifications to which document function the section is referring to). Created a new document for the Common Return Values: Process Results table (and organized the table to accommodate new columns).
1.40	Added detailed error codes for “Terminated with Error (error)”	Added codes SE91008~SE91xxx. * There is no need for modifications.
1.30	Updated specifications related to server maintenance	The following changes were made to server maintenance • Integrated error code SE10001 with SE10002. The old code SE10001 is discontinued. * If you are not using the error codes in your system (e.g. only showing the message “Under Maintenance”), there is no need for modifications. • Maintenance information is returned as a return value * Unless you wish to notify users of the contents of the maintenance outside of the screen-related functions (i.e. document and facial recognition functions), there is no need for modifications.
	End function “endVerify()” is now discontinued	We have discontinued the end function. * There is no need for modifications. endVerify() is deprecated but still remains as an empty function. Therefore, there is no need for modifications on your end.

1.21	Updated specifications related to server maintenance	The following changes were made to server maintenance • Integrated error code SE10001 with SE10002. The old code SE10001 is discontinued. * If you are not using the error codes in your system (e.g. only showing the message “Under Maintenance”), there is no need for modifications. • Maintenance information is returned as a return value * Unless you wish to notify users of the contents of the maintenance outside of the screen-related functions (i.e. document and facial recognition functions), there is no need for modifications.
	End function “endVerify()” is now discontinued	We have discontinued the end function. * There is no need for modifications. endVerify() is deprecated but still remains as an empty function. Therefore, there is no need for modifications on your end.
1.20	Started accepting Driving History Cards	The document function now accepts Driving History Cards. * There is no need for modifications if you do not accept this ID in your system.
	Started accepting passports	The document function now accepts passports. * There is no need for modifications if you do not accept this ID in your system.
	Added error code for “Terminated – Fatal Error”	Added error code SE94007. * There is no need for modifications.
	Added error code for “Unsupported Device”	Added an error code “Unsupported Device” for Android. * This error only occurs in document and facial recognition functions and we have prepared a dialog for it. However, if needed, please implement additional error handling. * Thus far we have confirmed that Android 602KC (DIGNO G) triggers this error.
	Available on new OS	The SDK is now available for iOS14 and Android11. * There is no need for additional implementations or modifications.
1.12	Common Return Values: Process Results	• Clarified which functions trigger dialogs • Changed the cause of error “Terminated – Fatal Error (SE90002)” to Undisclosed. • Added a note that the error “Unsupported Device” has not been observed thus far.
	5. SDK's Features	Added a bullet point on ID selection.
	6. Customizable Areas	Removed the section on adding a feature that allows customization of messages. * There is no need for additional implementations or modifications.
	7. Screen Content The order of the document and face scan images	Removed detailed images. * There is no need for additional implementations or modifications.
	9. Interface	Added a section on the prototype codes that will be available in the future. * There is no need for additional implementations or modifications.
1.11	Made corrections to the document function and facial recognition function: The match score could be negative	• The match score of facial image and ID image: × Positive integer ○ Integer of range -1000 to 1000 • The match score of facial image and live verification: × Positive integer ○ Integer of range -1000 to 1000

		* Modifications required: please adjust your implementations to accept negative values.
1.10	Common Return Values: Process Results	Clarified the return values each function can give.
	Modified the order of face scans	Changed the order of flash verification and facial muscle verification(now it is facial muscle verification→ flash verification). * There is no need for additional implementations or modifications.
	Updated Screen Capture	Updated the screen capture related to document and face scan * There is no need for additional implementations or modifications.
1.01	The permission denied dialog	Corrected the document since the SDK does give a dialog when permission is denied. * There is no need for additional implementations or modifications.
	Header font color settings	New feature * There is no need for modifications unless you want to change the font color.
1.00	[JP] Initial Version	

Summary of Document

This document contains the specifications of LIQUID eKYC Applicant SDK.

Service Overview

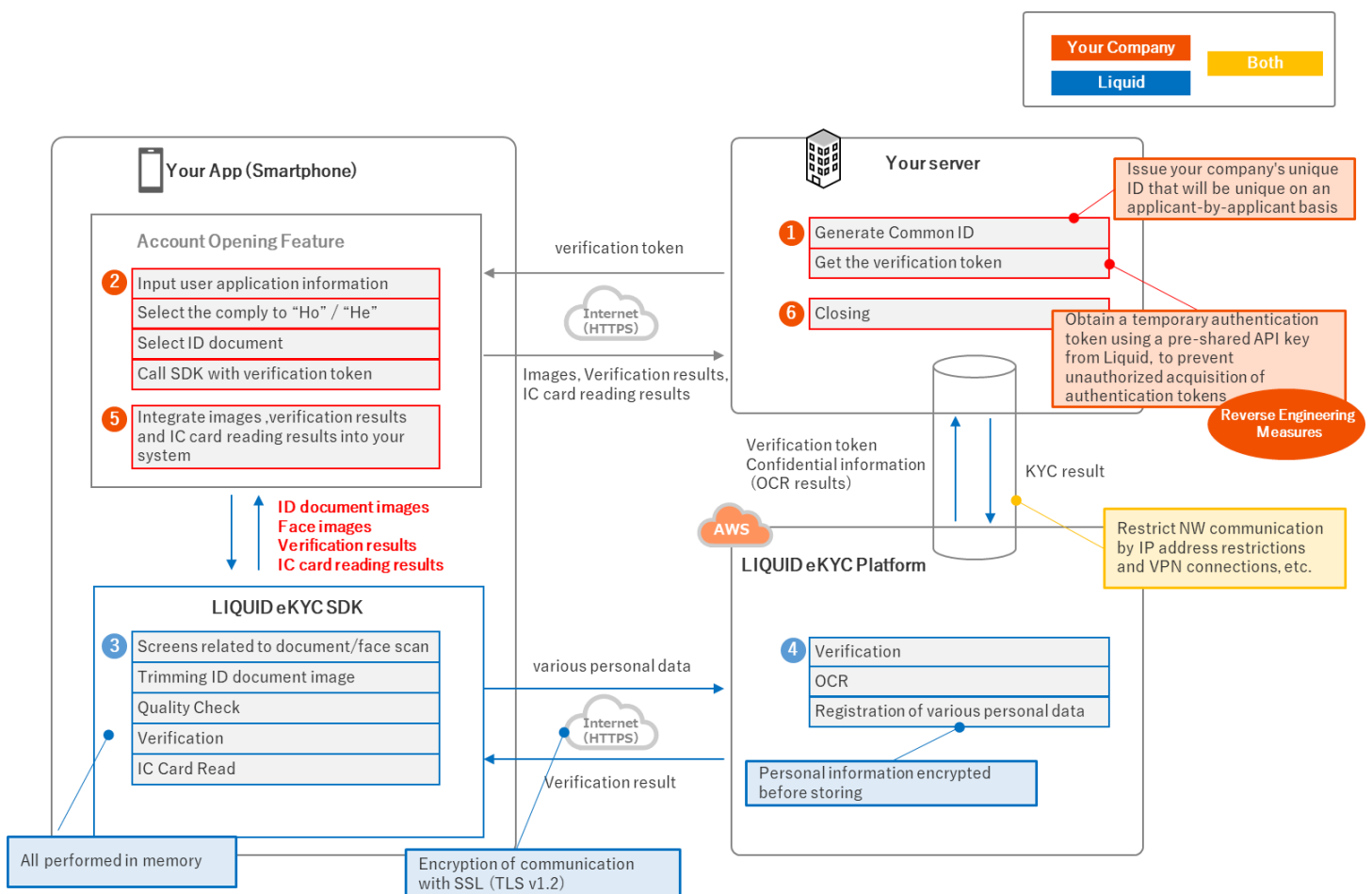
Below is the overview of “Liquid’s Facial Verification Service (SDK)”.

The service provides three methods for scanning and verifying images through the SDK. (One of which is for the trial version)

1. Method 1: Receive on Your Organization’s App

In this method your organization’s app will receive the scanned images and verification results from the SDK as return values.

Pros	The server implementations will be relatively simple since most features can be built into your app. * Processes related to verification tokens and end process must be linked with your server
Cons	Your app will need reinforced security because it will need to hold the user information.



* ① to ④ must be processed in 90 minutes at the latest (i.e. the verification token expires in 90 minutes).

* The results of the OCR process may be retrieved approximately 3 seconds after the process starts.

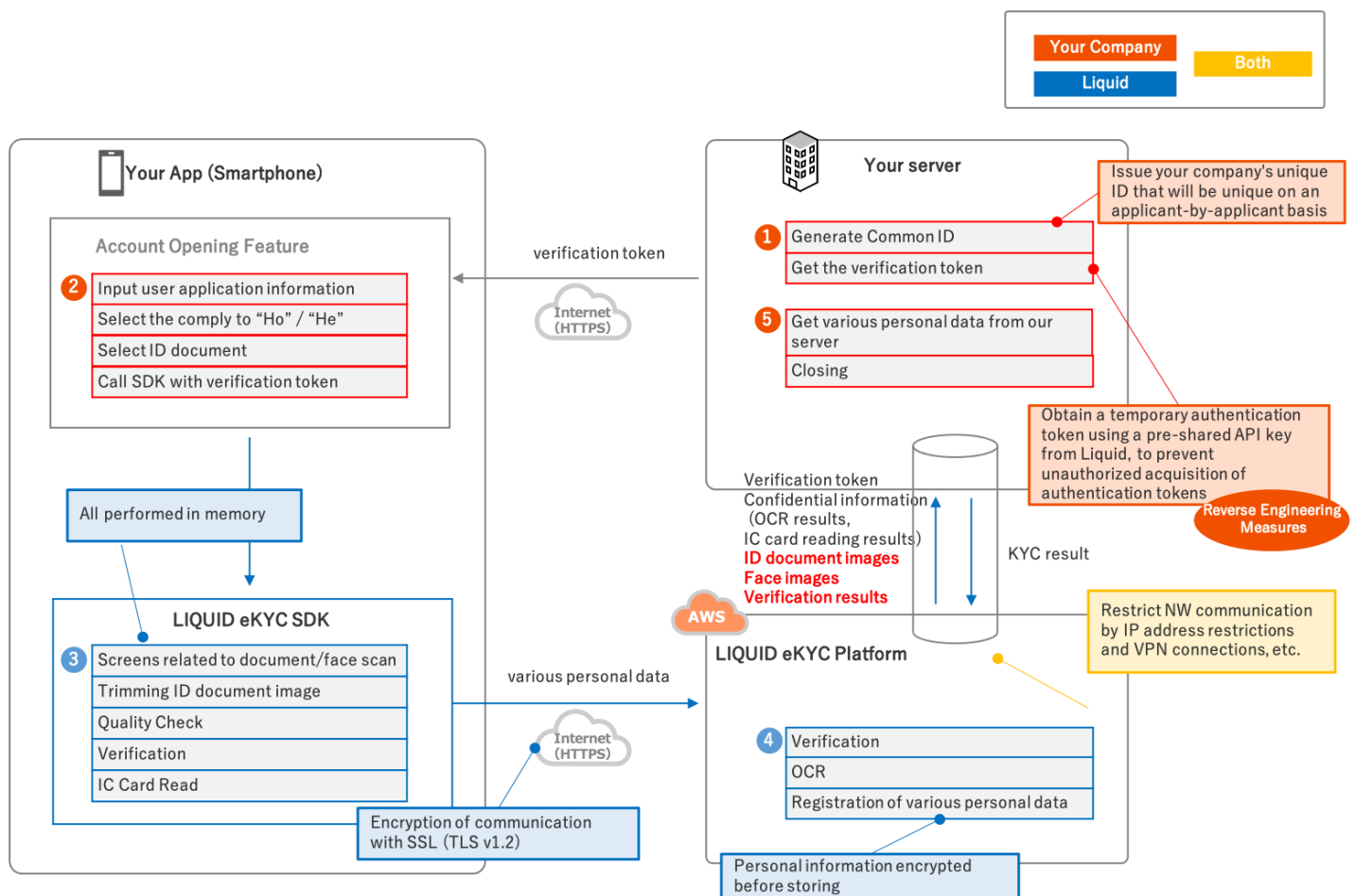
2. Method 2: Receive from Liquid's Server

In this method you will receive the scanned images and verification results via a connection established between your system and Liquid's server.

We have developed this method to meet the following demands:

- When your organization wishes to use the Web version and SDK version in parallel, and also wishes to unify the integration flow of sensitive data without using the Manager.
- When you wish to avoid image registration from fake apps (e.g. Phishing).

Pros	No additional security implementations needed since the user information in our platform is protected by a hash-based security that only allows registration from our official SDK (used when transmitting "various personal data" in the diagram below).
Cons	Server integration is required on your end to receive sensitive information.



* ① to ④ must be processed in 90 minutes at the latest (i.e. the verification token expires in 90 minutes).

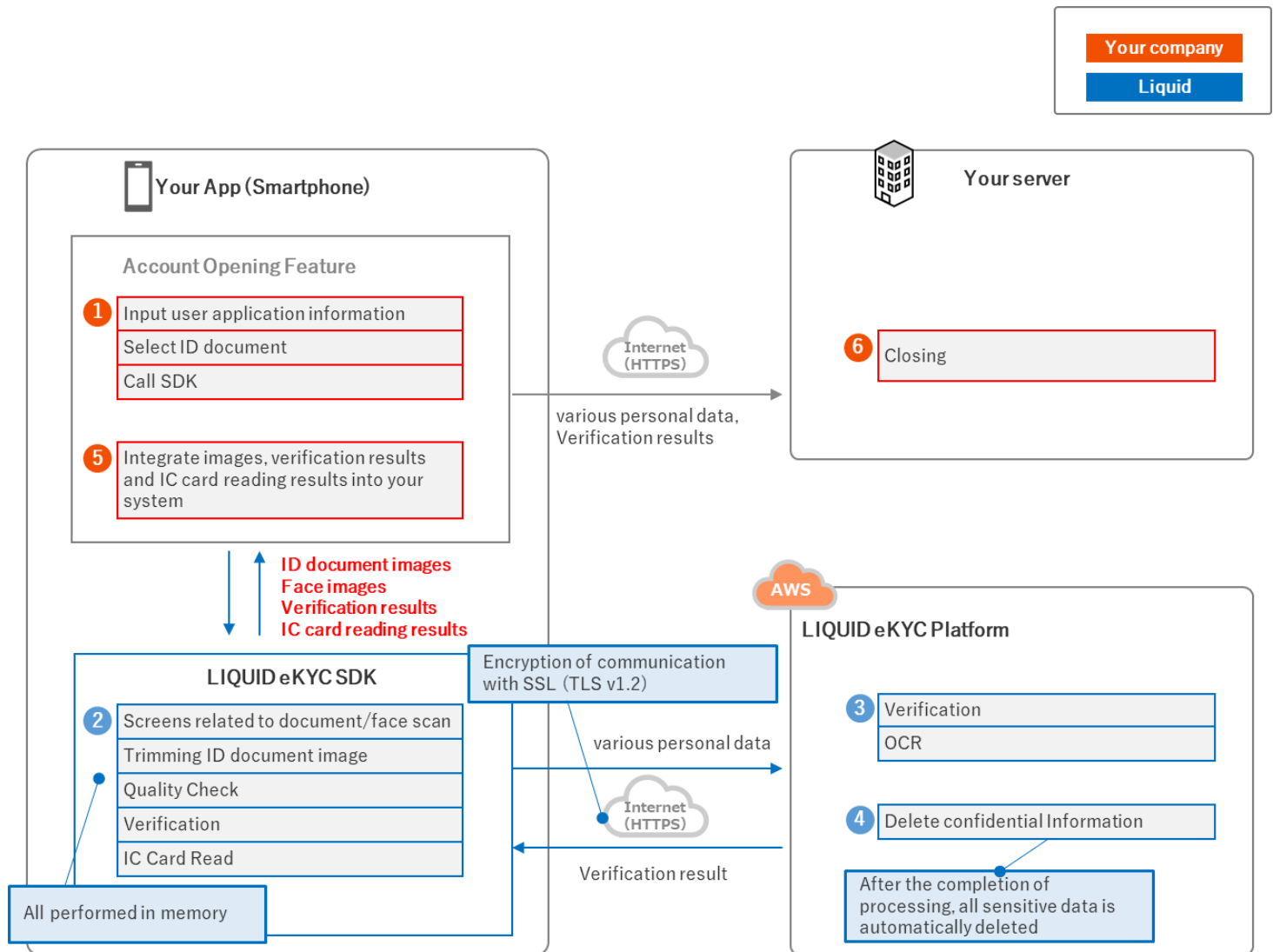
* The results of the OCR process may be retrieved approximately 3 seconds after the process starts.

Hash-Based Security

Hashes created by SDK's original algorithm are compared to the hashes received by the server to prevent unauthorized registration or modification of images from outside the SDK.

3. Method for Trial Version

The scanned images and verification results obtained through SDK will be sent as return values from the SDK. They will need to be received through your organization's app and synced with your system.

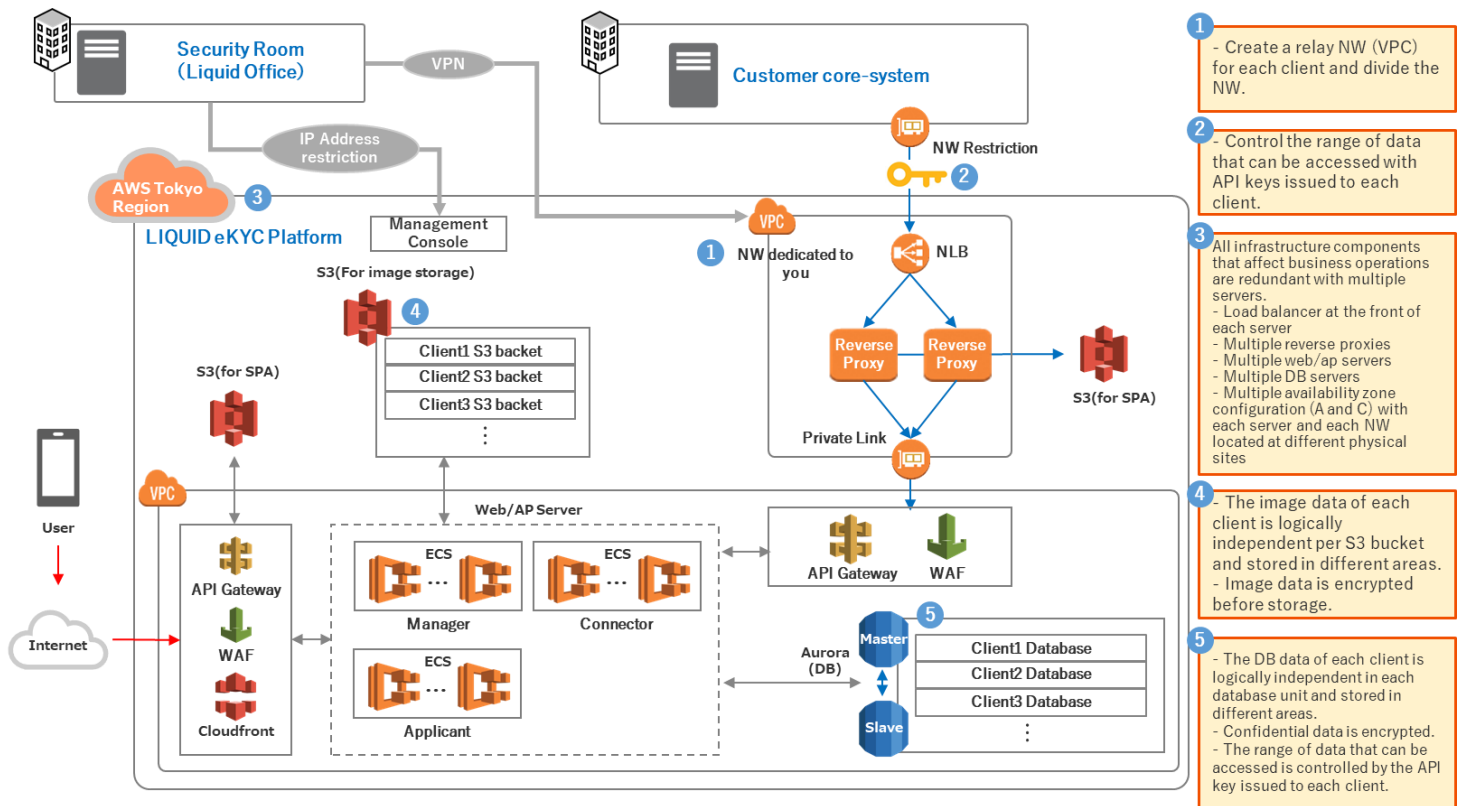


*① to ④ must be processed in 90 minutes at the latest (i.e. the verification token expires in 90 minutes)

*The results of the OCR process may be retrieved approximately 3 seconds after the process starts.

LIQUID eKYC Platform's Architecture in Detail

Shown below is LIQUID eKYC Platform's infrastructure map.



Operating Environment

The SDK operates on the environments below:

* The latest generation will be counted as the "first generation".

- The last 3 generations of iPhone
 - Development Language: Swift (Available in Swift 5) , Objective-C
 - IC scan Function: iPhone7 below (However, Driver's Licenses issued around the year 2019 and later cannot be read by iPhone7/7Plus)
 - ✦ OCR-Based Input Assistance Function for Authentication Information : iOS16.0 or later
 - Apple Wallet : iPhone Xs or later and iOS 18.5 or later (Note: When using Apple Wallet, the app cannot be installed on iOS versions below 16 (iOS 15 or earlier))
 - The last 7 generations of Android
 - Development Language: Kotlin, Java
 - IC scan Function: NFC Type B compatible devices only
 - Linkage processing with My number portal application (*): Android 9 or later (but only for devices with GP-SE)
- * Public Certification (smart phone JPKI) function is applicable

SDK's Features

The SDK provides the following features:

1. Comply to “Ho”

- Scanning ID documents
- Face scan of user

* The scanning process includes the actual scan (photoshoot) and verification of the images.

However, it does not include the following processes, which need to be created and managed by our clients implementing the system.

- ID selection screen
- Displaying instructions
- Confirmation screen for scans

2. Comply to “He”

- Reading IC cards of ID documents, (including signature verification)
 - * For Residence Cards/ Special Permanent Resident Certificate only, scanning the back of card
- Face scan of user

* The scanning process is approximately the same as Comply to “Ho”.

It also includes additional instructions for scanning the back of a Residence Card/ Special Permanent Resident Certificate.

* The IC card reading process includes the scan, verification, and displaying the instruction screens.

However, it does not include the following processes, which need to be created and managed by our clients implementing the system.

(We ask our clients to prepare these screens)

- ID selection screen
- Confirmation screen

■ About LIQUID eKYC's “Comply to ‘He’”

“Comply to ‘He’” is an alternative eKYC method that replaces the card scans (photoshoots) with reading the information on IC chip cards, such as Driver's Licenses. In this method the user will need to read the card chip with their smartphone and match that information with their face scan (selfie).

The greatest strength of the Comply to “He” method is that it does not require operators to check the scans as long as conditions are met. This reduces the process workload and allows real-time feedback of the eKYC results.

Clients using eKYC Manager can get the full benefits of this, as it automatically changes the eKYC status to “complete” when the user’s application meets the requirements. **Clients not using eKYC Manager** will first need to decide whether they want to skip the operator verification so they can receive the necessary information to do so.

* The requirements for skipping the verification are shown in the sequence overview

If the name or address on the Driver’s License has been changed, either the old or new name/address can be selected.

However, please note that there is no e-signature associated with the change.

* The "Retrieve Post-Change Name/Address on Driver's License" must be set in advance.

"Retrieve updated information" is selected: A driver's license with the updated information is available.

"Do not retrieve changed information" is selected: Driver's licenses with changed information will generate an error and processing will not continue.

For more details, please refer to the ‘LIQUID eKYC Applicant SDK Version Specification Additional Explanatory Material (License)’.

3. Comply to He (My Number)

- Function to obtain a My Number at the same time as the comply to He.

*Specify ‘National Identity Card (+My number)’ as the identification document type when executing the comply to He

4. Comply to "He" (Licence information)

- In addition to the Comply to "He" method, it is a function that enables simultaneous retrieval of license information embedded in the National Identity Card (commonly referred to as the "MyNa Driver's License").

※ Please specify “National Identity Card (+ License information)” as the ID document type.

5. Comply to “To”/”Chi” (IC Scan through SDK)

- Reading IC cards of ID documents (including signature verification)
 - * For Residence Cards/ Special Permanent Resident Certificate only, scanning the back of card

* The same as Comply to “He” but without the face scan

* It is also possible to simultaneously retrieve license information embedded in the National Identity Card (the so-called "MyNa Driver's License"). In this case, please specify "National Identity Card (+ License information)" as the ID document type.

* Since the eKYC SDK does not support the login confirmation of existing financial accounts as well as the confirmation by non-forwardable mail, "the check that the customer is on the confirmation record" must be performed separately when using the comply to "To", and "Confirmation by non-forwardable mail" must be performed separately when using the comply to "Chi".

6. Comply to "To"/ "Chi" (Photoshoot through SDK)

- Scanning ID documents

* The same as Comply to "Ho" but without the face scan

* Since the eKYC SDK does not support the login confirmation of existing financial accounts as well as the confirmation by non-forwardable mail, "the check that the customer is on the confirmation record" must be performed separately when using the comply to "To", and "Confirmation by non-forwardable mail" must be performed separately when using the comply to "Chi".

7. Front (Face)

- Photoshoot processing of Front ID documents
- Face photoshoot processing.

* Differences with Comply to "Ho" are as follows:

- Front side only (no diagonal or back side required)
- The photoshoot frame is fixed (doesn't move randomly)

8. Front (No-Face)

- Photo processing of the front of ID documents

* With the exception of the face photoshoot, it is consistent with the "Front photoshoot" method (with face photo).

9. FrontBack (Face)

- Photo processing of the front of ID documents
- Face photoshoot processing.

* The difference between the Comply to "Ho" is as follows.

- Only the front of the document is captured (other sides are not captured)
- The document's photo capturing frame is fixed (the frame will not move randomly)

10. FrontBack (No Face)

- Photo processing of the front of ID documents

* Same as the front photo method (face), except the face photo is not included.

11. Non-legal IC Scan (no e-signature verification, with facial verification)

- The same functionalities as Comply to “He” but without signature verification
- Reading IC cards of ID documents, **without** signature verification
 - * For Residence Card/ Special Permanent Resident Certificate only, includes scanning the back of card
- Face scan of user

* Exactly the same as Comply to “He”, but does not include signature verification

* **This does not comply with The Act on Prevention of Transfer of Criminal Proceeds because it excludes signature verification**

12. Non-legal IC Scan (no e-signature verification, without facial verification)

- The same functionalities as Comply to “To”/“Chi” (IC Scan through SDK) but without signature verification
- Reading IC cards of ID documents, **without** signature verification
 - * For Residence Card/ Special Permanent Resident Certificate only, includes scanning the back of card

* Exactly the same as Comply to “To”/“Chi” (IC Scan through SDK), but does not include signature verification

* **This does not comply with The Act on Prevention of Transfer of Criminal Proceeds because it excludes signature verification**

* It is also possible to simultaneously retrieve license information embedded in the National Identity Card (the so-called "MyNa Driver's License"). In this case, please specify “National Identity Card (+ License information)” as the ID document type.

13. Non-legal IC scan system (without digital signature verification and without face photo, offline)

- Non-legal IC scan system (without digital signature verification and without face photo, offline) that can be used even if there is no communication environment when the function is executed

*Note that **there is no signature verification and therefore does not comply with the Prevention of Transfer of Criminal Proceeds Act.**

*Note that **expiry dates are not checked.** Please verify the validity of the expiration date using the server time, not the device time.

14. Comply to “Wa” (Public Certification Service for Individuals)

- Reading IC cards on Individual Number Card (including Public Certification Service for Individuals)
- Linking with the Mynaportal application (including Public Certification Service for Individuals)

Note that there are two methods of JPKI.

- (1) Methods that do not involve the Mynaportal application
- (2) Methods that involve the use of the Mynaportal application

* Comply to "Wa" does not require photoshoots.

* The IC card reading process includes the scan, verification, and displaying the instruction screens.

However, it does not include the following processes, which need to be created and managed by our clients implementing the system.

(We ask our clients to prepare these screens)

- ID selection screen
- Confirmation screen

If you need to customize a series of processes such as the explanation screen, use the "JPKI Function (without screen)" which provides only the chip reading functionality.

Also, please note that this feature requires saving the verification results (For details, please refer to the JPKI Function section).

■ About LIQUID eKYC's "Comply to 'Wa'"

"Comply to 'Wa'" allows users to skip all photoshoots and verify themselves through their Individual Number Cards (feature unavailable for Driver's Licenses and Residence Card/ Special Permanent Resident Certificate). This feature uses the Public Certification Service for Individuals provided by the Japan Agency for Local Authority Information Systems (J-LIS).

The greatest strength of the Comply to "Wa" method is that it does not require operators to verify the applications that passed the Public Certification. This reduces the process workload and allows real-time feedback of the eKYC results.

15. JPKI + (Face)

- In addition to the Comply to "Wa"(JPKI), the function includes a process for capturing the face.

*Please make sure the face capturing is completed before proceeding with the Comply to "Wa" (JPKI).

*Please specify "JPKI+ (Face)" as the identity verification method when executing the comply to "Wa" (JPKI).

*While the Comply to "Wa" does not require the face capturing process, the following screen is not the same as it is for the Comply to "Ho", so it must be managed by the client's own system.

(These screens will need to be prepared by the client in advance)

- Instruction and other explanation screen

16. JPKI+ (My number)

- In addition to Comply to "Wa" (JPKI), function to obtain My numbers at the same time

*When executing "Comply to "Wa" (JPKI), please specify "National ID card (+ My number)" as ID document type.

17. JPKI+ (Face, My number)

- "JPKI+ (Face)" and "JPKI+ (My number)" functions in combination

*Please refer to the respective descriptions of "JPKI+ (Face)" and "JPKI+ (My number)"

18. JPKI + (Licence information)

- In addition to the Comply to "Wa" method (JPKI), it is a function that enables simultaneous retrieval of license information embedded in the National Identity Card (the so-called "MyNa Driver's License").

* Please specify "National Identity Card (+ License information)" as the ID document type.

19. Comply to "Ru" (card-substitute electromagnetic record)

- Read processing of National Identity Card information registered in Apple Wallet on iPhone (including digital signature verification)

※ The Comply to "Ru" method does not require any photoshooting processes.

※ The screen provided only displays the biometric authentication screen required to read the registered National Identity Card information from Apple Wallet; it does not include any screens for guidance or completion screens before or after that, etc. Therefore, screens such as those shown below should be prepared within the client's system.

- Screen for selecting the eKYC method and type of ID document
- Guidance screen related to Apple Wallet
- Result Confirmation Screen

Note that there is an obligation to retain the verification results (see Explanation of Card-Substitute Electromagnetic Record Function).

■ What is the comply to "Ru" in LIQUID eKYC?

The Comply to "Ru" method is a form of KYC verification that bypasses the physical National Identity Card scanning by reading the information on the National Identity Card registered in the iPhone's Apple Wallet. As a result, KYC verification can be completed using only a smartphone, even without having the physical National Identity Card on hand.

Specifically, since the National Identity Card information obtained from Apple Wallet is encrypted, decryption and digital signature verification are performed to confirm that the National Identity Card information has not been falsified or tampered with.

It should be noted that the Comply to "Ru" method is a distinctly different form of KYC verification from the Comply to "Wa" method (JPKI). Therefore, the functions of status confirmation and acquisition of the latest Basic Four Information that are under continuous customer management, which are available in the Comply to "Wa" method (JPKI), are not available.

The most notable feature of the Comply to "Ru" method is that KYC verification is completed using biometric authentication alone. With the Comply to "Wa" method (JPKI), acquiring National Identity Card information requires memory-based authentication credentials. In contrast, the Comply to "Ru" method only requires the user's smartphone and biometric data (Face ID facial recognition or Touch ID fingerprint authentication), which is expected to result in a lower abandonment rate. The second distinguishing feature is that the retrievable information includes a face photo (whereas the Comply to "Wa" method only provides the Basic Four Information). Finally, as with the Comply to "Wa" method (JPKI), when the result is normal, manual screenings by operators are no longer required, resulting in a significant reduction in operational costs and enabling real-time feedback of the KYC verification result to the user.

■ Design guidelines for screens to be provided by the client

The Digital Agency has published design guidelines for screens displayed to end users when requesting identity verification using a National Identity Card on iPhone.

Please refer to said guidelines for the screens to be developed.

20. Comply to "Ru" + Face

- In addition to the Comply to "Ru" method (card-substitute electromagnetic record), this function includes both the process of photographing the individual's face and the process of determining facial identity matching.

※ Please specify "Comply to 'Ru' + face" as the KYC verification method when performing the card-substitute electromagnetic record function.

※ The following screens are not provided for the Comply to "Ru" + face, and so should be prepared within the client's system.

(We kindly request our clients to prepare these screens in their system themselves.)

- Instructions and other explanation screens

■ What is the comply to "Ru" + Face in LIQUID eKYC?

The Comply to "Ru" + Face is a method of KYC verification that, in addition to retrieving National Identity Card information from Apple Wallet, includes photographing the individual's face. Specifically, in addition to the normal processing of the Comply to "Ru" method, it also determines whether the "face photo included in the acquired National Identity Card information" and "the individual's face photo taken on the spot in real time" are identical.

There are two main advantages to additionally taking a face photo:

The first is that, **once the National Identity Card has been registered in Apple Wallet**, the user no longer needs to carry the physical card at all times. **Compared to KYC verification methods that require reading the physical National Identity Card (such as the Comply to "He" or "Wa" methods), it is assumed that the card can be more easily obtained through illicit means such as monetary exchange, coercion, or fraud.** Since the physical card can be returned to the victim after it has been registered in Apple Wallet through fraudulent activity, the psychological barrier for the victim may be lower, or they may not realize that the registration occurred under false pretenses. The Comply to "Ru" + face method can prevent such a National Identity Card fraudulently registered to Apple Wallet from being used by matching it with the face photographed on the spot.

The second advantage is that facial recognition can be used for personal authentication in ongoing service use after an account is created. Since the face information contained in a National Identity Card is a black-and-white, low-quality image, it is not suitable for highly accurate personal authentication. With the Comply to "Ru" + Face method, a colour face photo of the individual is taken for authentication purposes, enabling the implementation of facial recognition in future personal authentication scenarios.

■ Design guidelines for screens to be provided by the client

The Digital Agency has published design guidelines for screens displayed to end users when requesting identity verification using a National Identity Card on iPhone.

Please refer to said guidelines for the screens to be developed.

21. Comply to "Ru" + (My Number)

- A function to obtain a My Number, as with the Comply to "Ru" method

※When executing the Comply to "Ru" method (JPKI), please specify "National Identity Card (+ MyNumber)" as the ID document type.

22. Comply to "Ru" + (Face and My Number)

- A function that combines both 'Comply to "Ru" + (Face)' and 'Comply to "Ru" (+ My Number)'

※Please refer to the respective explanations for 'Comply to "Ru" + (Face)' and 'Comply to "Ru" (+ My Number)'

23. Additional Document Photo

- A function to photograph documents used for tasks other than KYC verification

※"My Numbers" must not be obtained from additional documents

※This function can be used to photograph income certificates required for loan screening, health certificates required for health insurance screening, etc. There are no restrictions on the documents that can be photographed.

※Must be used in conjunction with functions that handle IC cards, such as Comply to "He" method. Cannot be used in conjunction with functions for taking photographs, such as Comply to "Ho" method.

24. Face Verification

- Face Verification Processing

Receives two facial images and performs verification processing on the LIQUID eKYC Platform.

The facial images to be processed are as follows:

- Facial image on the document surface
- Facial image captured as a selfie

The processing results are as follows:

- Facial photo verification result (High / Medium / Low)
- Match score

※For details of the interface, please refer to the "LIQUID eKYC Applicant SDK Interface Specifications"

※Please note that this function does not include any screens or capture functionality.

(Important) Recommended sequence

The LIQUID eKYC SDK integration method and processing sequence are organized in the file below.

Please be sure to read the Liquid recommended flow as well.

Recommended Sequence for Compliance with the Act on Prevention of Transfer of Criminal Proceeds

eKYC method	File name of sequence diagram
Overall, comply to "Wa"	[EN]Overall_Comply to Wa_v20251203.pdf
Overall, JPKI+(Face)	[EN] Overall_JPKI+(Face)_20251203.pdf
Overall, comply to "He"	[EN]Overall_Comply to He_ 20251203.pdf
Overall, comply to "Ho"	[EN]Overall_Comply to Ho_v20240408.pdf
Without screen, comply to "He"	※Please contact our sales team for more information
Without screen, JPKI+ (MyNumber)	※Please contact our sales team for more information

Recommended sequence for each function

eKYC method	File name of sequence diagram
Comply to "Ho"(evaluation version)	[EN]SDK Comply "Ho"(evaluation version) v20250327.pdf
Overall_Comply to Wa_(Evaluation version)	[EN]Overall_Comply to Wa_(Evaluation version) v20260109
Comply to "To"/"Chi" (IC Scan through SDK(Evaluation version))	[EN]Comply to "To"/"Chi" (IC Scan through SDK(Evaluation version)) v20260109
Comply to "To"/"Chi" (Photoshoot through SDK)	[EN] Comply to ToChi(Photoshoot through SDK) v20240822.pdf
Front (Face)	[EN] Front (Face) v20240408.pdf
Front (No face)	[[EN] Front(No face) v20240408.pdf
Comply to "To"/"Chi" (IC Scan through SDK)	[EN] SDK Comply ToChi IC Reading Basic Sequence v20250327.pdf
Non-legal IC scan system (without digital signature verification but with face photo)	[EN]SDK Non-legal IC scan (Non-Verification, Face) Basic Sequence v20250327.pdf
Non-legal IC scan system (without digital signature verification and without face photo)	[EN]SDK Non-legal IC scan (digital signature verification and No face photo)_ v20250327.pdf
Comply to "Ru"、 Comply to "Ru" + Face	[EN]Comply to Ru (card-substitute electromagnetic record) v20251112.pdf
【Liquid specified. For special purpose operators】 Face Verification Standalone Function	※ Please contact our sales team for more information.

Customizable Areas

The SDK's UI is designed to be simple and easy to understand. Normally it can be used without customizations. However, the below areas may be customized depending on our clients' needs.

- Button Color
- Text color at the top of the shooting screen
 - e.g., Text color of "driver's license, front side"

Screen Content

Screen images

For SDK screen images, please refer to "eKYC Applicant (SDK version) Product Specification".

The Order of Functions

The order of the functions explained above can be changed.

For example:

- <Comply to "Ho"> document scan → Capturing face
- Capturing face → <Comply to "Ho"> document scan
- <Comply to "He"> IC card scan → Capturing face
- <Comply to "He"> IC card scan → (stop the <Comply to "He"> process) → <Comply to "Ho"> document scan → Capturing face

- <Comply to “Wa”> IC card scan → (stop the <Comply to “Wa”> process) → <Comply to “He”> IC card scan → Capturing face

For JPKI+ (face), the function execution timing is fixed as shown below, therefore please complete the face capturing first.

- Capturing face → JPKI

However, only for the Comply to "Ru" + face method, the timing of execution of the function is fixed as follows, so please ensure that the face is photographed afterwards.

- Apple Wallet reading (card-substitute electromagnetic record) → Photographing the individual's face

Recommendations

If you are going to use LIQUID eKYC Applicant's OCR feature, the document scan must be done before the user puts in personal information (name/date of birth/address).

Additionally, in that case we recommend setting the flow as document scan → face scan → personal information (name/date of birth/address) because OCR takes some time to process. This way the user will not have to wait for the OCR to finish.

How The Integration Should Work

LIQUID eKYC Applicant SDK runs as an integrated application within our client's app.

Below are functions/arguments and corresponding ID Document type codes for using each feature.

* About ID Document type codes

They are codes assigned to ID documents that are used for KYC in eKYC Connector (the external linkage API).

For details, please refer to the specifications for Connector. These are mainly used by clients with eKYC Manager to retrieve KYC results.

eKYC Method	Corresponding functions/arguments
Comply to "Ho"	- Initialization Function - Terms of Use Function - Document Function Argument: eKYC Method=COMPLY_HO - Facial Recognition Function - Activate Function
Comply to "To" (Photoshoot through SDK)	- Initialization Function - Terms of Use Function - Document Function Argument: eKYC Method=COMPLY_TO - Activate Function
Comply to "Chi" (Photoshoot through SDK)	- Initialization Function - Terms of Use Function - Document Function Argument: eKYC Method=COMPLY_CHI - Activate Function
Front (Face)	- Initialization Function - Terms of Use Function - Document Function Argument: eKYC Method=FRONT_FACE - Facial Recognition Function - Activate Function
Front (No-face)	- Initialization Function - Terms of Use Function - Document Function Argument: eKYC Method=FRONT - Activate Function
FrontBack (Face)	- Initialization Function - Terms of Use Function - Document Function Argument: eKYC Method= FRONT_BACK_FACE - Facial Recognition Function - Activate Function
FrontBack (No-Face)	Initialization Function

	• Terms of Use Function • Document Function Argument: eKYC Method= FRONT_BACK • Activate Function
Comply to “He”	• Initialization Function • Terms of Use Function • Document IC Card Function Argument: eKYC Method=COMPLY_HE • Facial Recognition Function • Activate Function ■ About MyNa Driver's Licenses ■ When a National Identity Card is specified as the KYC verification document type, a National Identity Card with registered license information (commonly referred to as the “MyNa Driver's License”) can be used for KYC verification without any issues. ※ However, the driver's license information will not be read. If license information is required, please use the "Comply to 'He' method + (license information)" instead.
Comply to “He” *without screen	• Initialization Function • IC card reading availability Function • Document IC Card Function (without screen) Argument: eKYC Method=COMPLY_HE • Facial Recognition Function • Activate Function ■ About MyNa Driver's Licenses ■ When a National Identity Card is specified as the KYC verification document type, a National Identity Card with registered license information (commonly referred to as the “MyNa Driver's License”) can be used for KYC verification without any issues. ※ However, the driver's license information will not be read. If license information is required, please use the "Comply to 'He' method + (license information)" instead.
Comply to “He” + (My Number)	• Initialization Function • Terms of Use Function • Document IC Card Function Argument: : ID document type=National identity card (+MyNumber) eKYC Method=COMPLY_HE • Facial Recognition Function • Activate Function

■ About MyNa Driver's Licenses ■

When a National Identity Card is specified as the KYC verification document type, a National Identity Card with registered license information (commonly referred to as the “MyNa Driver's License”) can be used for KYC verification without any issues.

※ However, the driver's license information will not be read. If license information is required, please use the "Comply to 'He' method + (license information)" instead.

Comply to "He" + (Licence information)	• Initialization Function • Terms of Use Function • Document IC Card Function Arguments : ID document type = National Identity Card (+License Information) eKYC method = COMPLY_HE • Facial Recognition Function • Activate Function
Comply to “To” (IC Scan through SDK)	• Initialization Function • Terms of Use Function • Document IC Card Function Argument: eKYC Method=COMPLY_TO • Activate Function
Comply to “Chi” (IC Scan through SDK)	• Initialization Function • Terms of Use Function • Document IC Card Function Argument: eKYC Method = COMPLY_CHI • Activate Function
Non-legal IC Scan (no e-signature verification, with facial verification)	• Initialization Function • Terms of Use Function • Document IC Card Function Argument: eKYC Method=READ_FACE • Facial Recognition Function • Activate Function
Non-legal IC Scan (no e-signature verification, without facial verification)	• Initialization Function • Terms of Use Function • Document IC Card Function Argument: eKYC Method=READ • Activate Function
Non-legal IC Scan (no digital signature verification, without facial verification, offline)	• Document IC Card Function Argument: eKYC Method=READ OFFLINE
Comply to “Wa”	• Initialization Function • Terms of Use Function • JPKI Function

	Argument: : ID document type= National identity Card eKYC Method=COMPLY_WA • Activate Function ■ About MyNa Driver's Licenses ■ When a National Identity Card is specified as the KYC verification document type, a National Identity Card with registered license information (commonly referred to as the "MyNa Driver's License") can be used for KYC verification without any issues. ※ However, the driver's license information will not be read. If license information is required, please use the "Comply to 'He' method + (license information)" instead.
Comply to "Wa" ※Smartphone JPKI	• Initialization Function • Terms of Use Function • Japanese Public Key Infrastructure (smartphone JPKI) Argument: : ID document type= National identity Card eKYC Method=COMPLY_WA • Activate Function
Comply to "Wa" *without screen	• Initialization Function • IC card reading availability Function • JPKI Function (without screen) Argument: : ID document type= National identity Card eKYC Method=COMPLY_WA • Activate Function ■ About MyNa Driver's Licenses ■ When a National Identity Card is specified as the KYC verification document type, a National Identity Card with registered license information (commonly referred to as the "MyNa Driver's License") can be used for KYC verification without any issues. ※ However, the driver's license information will not be read. If license information is required, please use the "Comply to 'He' method + (license information)" instead.
JPKI+ (MyNumber)	• Initialization Function • Terms of Use Function • JPKI Function Argument: : ID document type= National identity Card (+MyNumber) eKYC Method=COMPLY_WA • Activate Function
JPKI+ (MyNumber) ※without screen	• Initialization Function • IC card reading availability Function • JPKI Function (without screen)

	Argument: : ID document type= National identity Card(+MyNumber) eKYC Method=COMPLY_WA • Activate Function
JPKI+ (Face)	• Initialization Function • Terms of Use Function • Facial Recognition Function *Make sure this is executed before the JPKI function • JPKI Function Argument: ID document type= National identity Card eKYC Method=FACE_JPKI • Activate Function
JPKI+ (MyNumber、 Face)	• Initialization Function • Terms of Use Function • Facial Recognition Function *Make sure this is executed before the JPKI function • JPKI Function Argument: : ID document type= National identity Card (+MyNumber) eKYC Method=FACE_JPKI • Activate Function
JPKI + (Licence information)	• Initialization Function • Terms of Use Function • JPKI Function Arguments : ID document type= National Identity Card (+License Information) eKYC method = COMPLY_WA • Activate Function
Comply to "Ru"	• Initialization Function • Terms of Use Function • Card-Substitute Electromagnetic Record function Arguments : ID eKYC method = Comply to "Ru" (Apple Wallet) ID document type = National Identity Card • Activate Function
Comply to "Ru" + (My Number)	• Initialization Function • Terms of Use Function • Card-Substitute Electromagnetic Record function Arguments : ID eKYC method = Comply to "Ru" (Apple Wallet) ID document type = National Identity Card + (My Number) • Activate Function
Comply to "Ru" + Face	• Initialization Function • Terms of Use Function • Card-Substitute Electromagnetic Record function

	Arguments : ID eKYC method = Comply to "Ru" + Face (Apple Wallet) ID document type = National Identity Card • Facial Recognition Function ※ Please be sure to execute after the Card-Substitute Electromagnetic Record Function • Activate Function
Comply to "Ru" + Face (+My Number)	• Initialization Function • Terms of Use Function • Card-Substitute Electromagnetic Record function Arguments : ID eKYC method = Comply to "Ru" + Face (Apple Wallet) ID document type = National Identity Card (+My Number) • Facial Recognition Function ※ Please be sure to execute after the Card-Substitute Electromagnetic Record Function • Activate Function

Also refer to the specification documents (shown at the end of this document) for sequence diagrams and details.

* The systems related to the SDK are listed below.

- Our clients' app (containing the SDK)
Is the interface for users.
- LIQUID eKYC Applicant SDK (eKYC SDK)
Provides document scan and face scan features for the users.
- Our clients' system
Our clients' system, wherein the connection with LIQUID eKYC Connector is established.
- LIQUID eKYC Connector
Provides a secure connection between our clients' system and the Connector using verification tokens and other safety measures.
- LIQUID eKYC Manager * For clients using the version that includes Manager
For management-related operations, such as checking the scanned images.

Detail Comply to "Ho", Comply to "To"/"Chi" (Photoshoot through SDK)

No additional specifications

Detail Comply to "He", Comply to "To"/"Chi" (IC Scan through SDK)

Specifications for status auto-verification process

When the following requirements are met by clients using eKYC Manager, the eKYC status is automatically changed to "complete", and no operator verification will be needed.

Not eligible for auto-verification (every application needs to be checked by operator)

■ For Driver's License and Individual Number Card

Requirements	Content
Application information and IC card information matches	OK when the name, date of birth, and address recorded in the card match those of the application.
ID photo and the face scan matches	OK when the score is equal to or above the predefined value
Flash verification and the face scan matches	OK when the score is equal to or above the predefined value

■ For Residence Card/ Special Permanent Resident Certificate

Requirements	Content
Application information and scanning result of IC card information (Front) matches	*The IC chip in the Residence Card/Special Permanent Resident Certificate contains the card's surface image without text data. Therefore, OCR technology is used to extract text from the front images and the text information can be used to determine whether there is a match. *Backside images of Resident Card/Special Permanent Resident Certificate are not OCR compatible, so only the front side information is used to determine a match. *Therefore, if an end-user mistakenly enters the old address instead of the new address when the address is written on the reverse side of the application, the old address on the front side of the application will be judged as the same as the new address, and the application may be marked as OK and "completed". OK when the name, date of birth, and address recorded on the card (Front) match with the application.
ID photo and the face scan matches	OK when the score is equal to or above the predefined value
Flash verification and the face scan matches	OK when the score is equal to or above the predefined value

Tamper detection for IC card scan results

Use eKYC Connector's "LIQUID_eKYC_022_GetICCardInfo_API" for tamper-proof retrieval of IC chip card scan results.

Additionally, please note that using the return values from the SDK directly will make the data vulnerable to tampering during the linkage between the SDK and your organization's app or processing within your organization's app.

Handling new names and addresses in Driver's License

Tamper detection cannot be done for new names and addresses in a Driver's license because their information is not linked to the e-signature of the card. Therefore, our clients can change the master setting below according to their preferences.

Setting "Whether or not the changed name/address is obtained on the driver's license."	Specifications and operations
"Retrieve the information after the change".	New names and addresses are used. However, note that it is technically possible for someone to tamper with this data.
" Do not retrieve changed information"	New names and addresses are not used, hence data can be retrieved in a secure way through Connector. However, this will return an error when a new name or address is registered within the Driver's License's IC chip.

Data outside of the e-signature verification's scope

Tamper detection cannot be applied to some information on ID cards.

- Driver's Licenses
Other than the new name/address information stated above, external character images are out of the tamper detection's scope
- Individual Number Cards
The expiration date of the card is out of the tamper detection's scope
- Residence Cards/ Special Permanent Resident Certificate
Application for extension of period of stay, Residence Card/ Special Permanent Resident Certificate information type, Residence Card/ Special Permanent Resident Certificate type, Permission to Engage in Activity other than that Permitted under the Status of Residence Previously Granted are out of the tamper detection's scope

Detail Comply to "Wa"

Tamper detection for IC card scan results

If you wish to securely obtain the IC card scan results through the application without the risk of tampering, use the eKYC Connector "LIQUID_eKYC_022_GetICCardInfo_API".

On the other hand, if you use the returned values of the SDK as they are, please note that data tampering is possible in principle when linking from the SDK to your application and during the processing on your side.

Comply to "Ru" Method Details

Tampering detection of Apple Wallet reading results

To securely and reliably obtain National Identity Card information via Apple Wallet reading without the risk of data tampering, use the eKYC Connector "LIQUID_eKYC_022_ICCardReadDataAcquisitionAPI."

On the other hand, if you use the returned values of the SDK as they are, please note that data tampering is possible in principle when linking from the SDK to your application and during the processing on your side.

Detail Comply to “Ho” (Trial Version)

No additional specifications

Connection Methods with LIQUID eKYC Connector

Outline

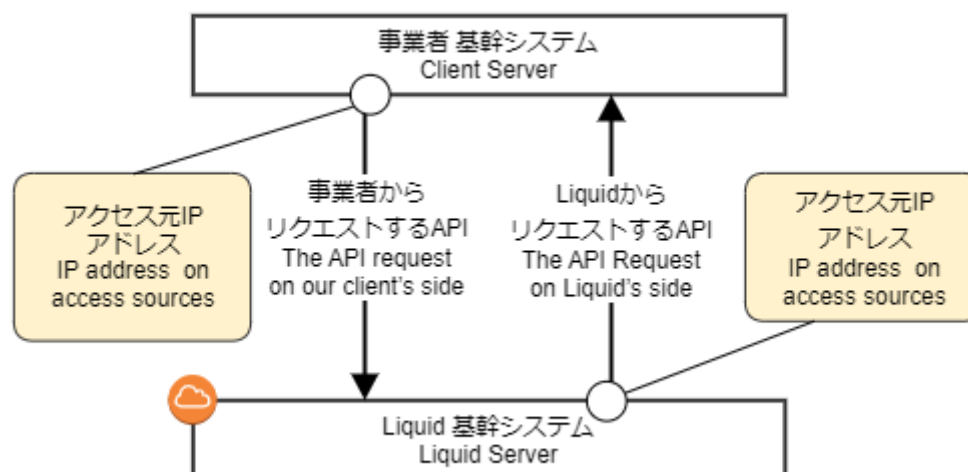
- In order to use eKYC Connector, our clients need to connect their system with LIQUID eKYC Connector using the methods shown below.
- The API needs some network restrictions for access, so end users cannot call the API directly from their devices.
- We accept methods other than “Connection using an IP address” on a separate fee.

For details, please contact our sales representative

Methods of connection

Method 1. Connection using an IP address

- The API request on our client’s side
 - Liquid employs IP address control on access sources.
 - LIQUID eKYC Connector’s IP address is dynamic. Therefore our clients cannot use IP address restriction in the connection.
- The API Request on Liquid’s side (Verification Result API)
 - This feature is added upon request.
 - Our clients employ IP address control on access sources.
 - If necessary, Liquid can add values and elements in the HTTP headers upon request.



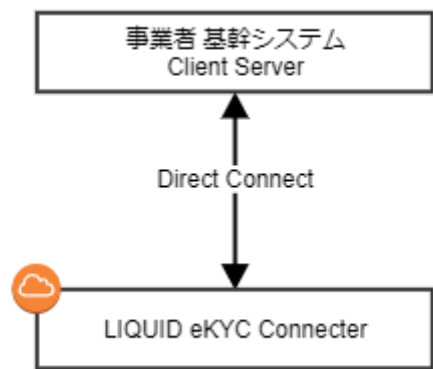
Method 2. Dedicated network connection with Liquid’s on-premise environment (AWS Direct Connect)

- Connection via a private network with dedicated network connection (Direct Connect).

For when our clients wish to establish a dedicated connection instead of a shared one using DirectConnect

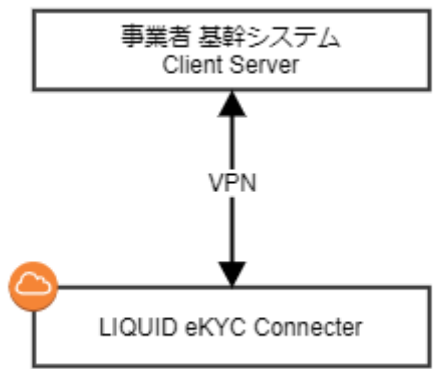
When using this method our clients are responsible for preparing an AWS account and requesting a LOA-CFA(Letter of Authorization and Connecting Facility Assignment).

Once the dedicated connection ready, a VIF will be needed to connect to the Liquid account.



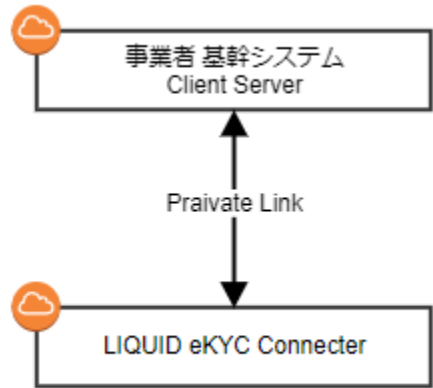
Method 3: VPN connection with Liquid’s on-premise environment

- Connection via a private network using VPN.



Method 4: Direct connection between Liquid’s and client’s AWS environment (AWS Private Link)

- Connection via a private network using Private Link.



SDK

When the following functions are used in the client's app,

- The document function
- The document ID Card function
- The JPKI function
- The facial recognition function

simply calling a function will activate and complete the document scan/IC card scan/verification.

However, clients need to remember that there are accompanying functions that may need to be addressed (please refer to the sequence overview).

For installation and integration of the SDK, please refer to the README file provided separately.

For names of functions and values, please refer to the API Documents provided separately.

The interface can be changed with prior notice.

As new parameters in the response may be added at a high frequency, they should be implemented in a way that will not affect existing processing.

Initialization Function “startVerify()”

This prepares other functions for use.

For details of the interface, please refer to the "LIQUID eKYC Applicant SDK Interface Specifications".

Initialization Function (for Trial Version) “startVerify()”

The initialization function in the trial version differs from the standard SDK.

This prepares other functions for use.

Please note that when calling this function, clients will need to prepare the following. This is necessary because the function activates communication between systems.

1. A UI that notifies the user about the data transmission (e.g. a loading screen).
2. A rerun of the function when the process fails (e.g. connection failures). For example, prompting the user to rerun through a dialog.

Liquid needs to activate settings for clients to use the trial version

Please contact our sales team for more information

For details of the interface, please refer to the "LIQUID eKYC Applicant SDK Interface Specifications".

Terms of Use Function “showTermsOfUse()”

A process that should be used after the Initialization Function.

Displays terms of use and privacy policy of LIQUID eKYC. The user needs to agree to these before using the other functions. If the user of that Applicant ID has agreed to the terms before, a dialog is shown to notify that before terminating successfully.

Note that an error will be returned if the activate function is executed without performing the relevant function and without obtaining consent.

- When the function returns “success”
 - Can use Document Function and other functions
- When the function does not return “success”
 - The user has not agreed to the terms of use and privacy policy. This function must be executed again

For details of the interface, please refer to the “LIQUID eKYC Applicant SDK Interface Specifications”

Document Function “verifyIdDocument()”

Scans and verifies documents.

A screen is also provided to check the photos taken (a screen where users themselves can check the photos taken and decide whether or not to re-shoot them).

*Use of the confirmation screen for photos taken is optional, but recommended.

For details of the interface, please refer to the "LIQUID eKYC Applicant SDK Interface Specifications".

Document IC Card Function “verifyIdChip ()”

Reads and optionally verifies IC cards (this is different from public certification). The e-signature verification may be turned ON or OFF. However, turning the e-verification OFF is not in compliance with the legal requirements of Comply to “He” and Comply to “To”/“Chi” (IC Scan through SDK)

Additionally, the OCR-based input assistance function for authentication information can be used.

	Android	iOS
Supported OS	Refer to "Operating Environment"	Refer to "Operating Environment" ※The OCR camera screen is not shown on versions earlier than iOS16.0. (Subsequent screens can be used as usual.)

Please note that external character handling is different for each document type.

- Driver's License: external characters may be returned (please refer to the interface details)
- Individual Number Card: no external characters are returned (since they are already replaced by JIS level one and two kanji and JIS supplementary kanji)
- Residence Card/ Special Permanent Resident Certificate: if there are external characters, they are shown on the card as an image

Precautions for each document

- For Residence Cards/ Special Permanent Resident Certificate, a photo of the back of the card must be taken
There is also a screen for reviewing the photos taken (where users can review the photos they have taken and decide whether or not to re-take them)
*The use of the screen to check the photos taken is optional, but recommended.
- Information such as names cannot be read from the chip inside Residence Card/ Special Permanent Resident Certificate.
In some cases, the information must be read through OCR using the image of the card recorded in the IC chip.
For details, please refer to "Connector OCR Result Retrieval API Specifications"
* The OCR of the Residence Card/ Special Permanent Resident Certificate in Comply to "He", Comply to "To"/"Chi" (IC Scan through SDK), and Comply to "Ho" are equivalent (can also retrieve resident status and period of stay)
- Individual Number Cards with special date of birth expressions cannot not be read.
e.g. 2020/A1/A3 (at the end of spring in 2020), 0000/01/00 (on an unknown year, January, on an unknown date)

For iOS, "AID information" needs to be specified on Xcode.

"AID information" is a fixed character string on Driver's Licenses, Individual Number Cards, and Residence Cards/Special Permanent Resident Certificate,

but for Individual Number Cards, a NDA must be signed for disclosure of "AID information".

** The NDA process will take around two to four weeks.*

Please contact our sales team for more information.

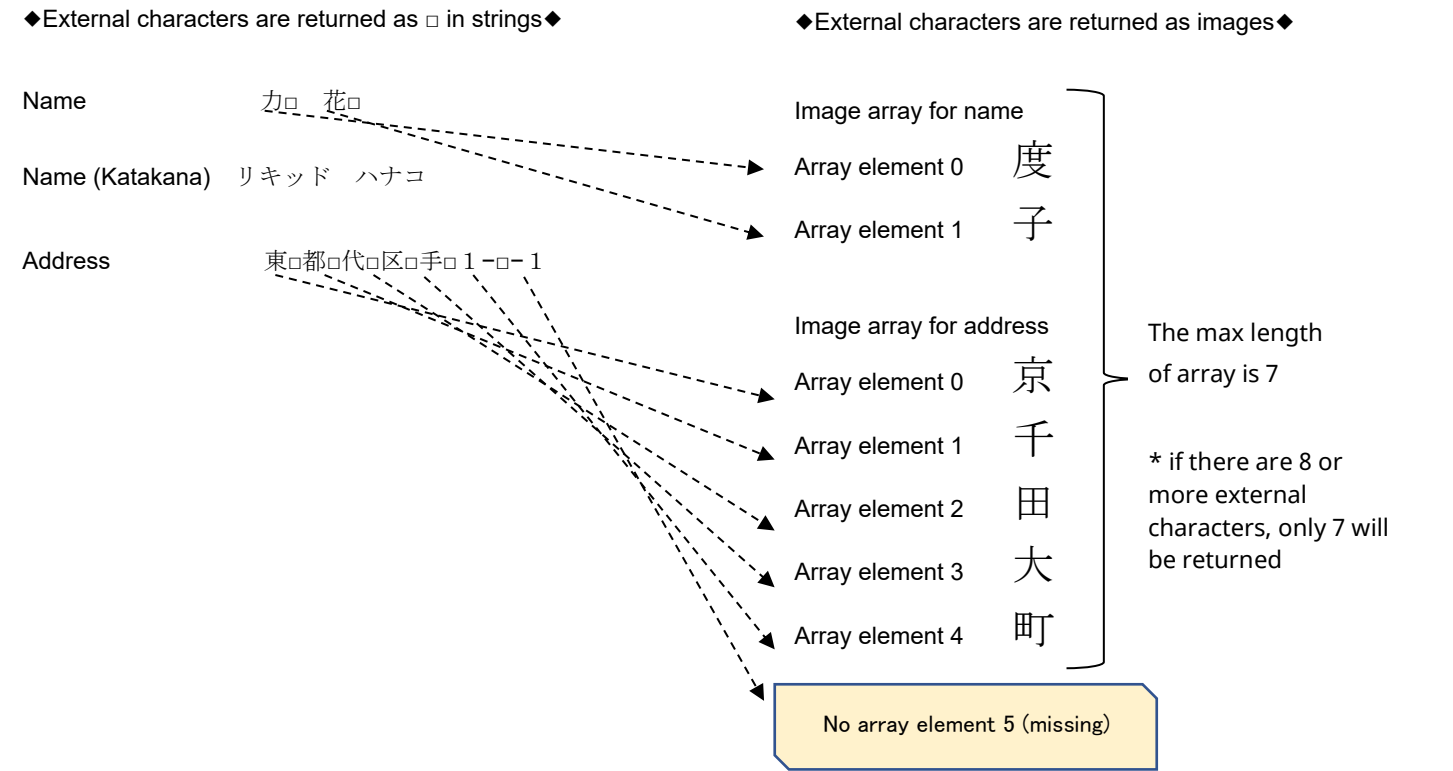
The external characters for Driver's License

Below is an example of what may be returned.

◆Let us pretend that the external characters are external characters◆

Name カ度 花子
Name (Katakana) リキッド ハナコ
Address 東京都千代田区大手町 1-6-1

* Please note that these are not actual external characters

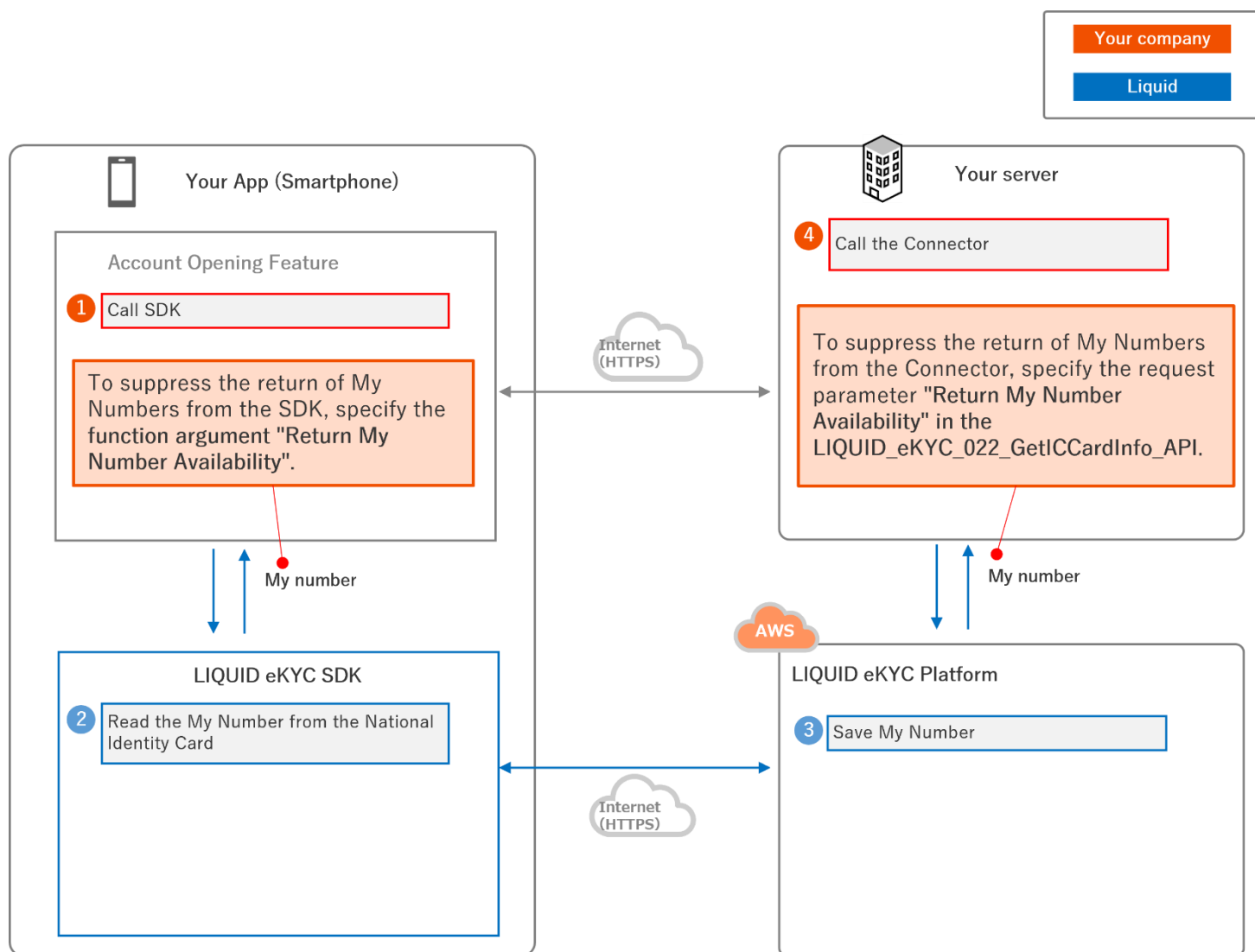


* There is no predefined rule on which character will be missing when there are more than 7 external characters (that is decided by the owner of the Driver's License).

*If the name or address is missing, a "missing external character image on IC chip" error is returned.

Control of My Number Acquisition for National Identity Cards

The return control method to the client's app or core system is as follows:



For details of the interface, please refer to the "LIQUID eKYC Applicant SDK Interface Specifications".

Document IC card function (without screen) "verifyIdChipNonInteractive ()"

The function of "Document IC card function" is retained and only the chip-reading part of the screen is provided.

The chip reading display (henceforth referred to as "half-modal") is superimposed on the screen of the client's app, and communication takes place asynchronously.



Therefore, the following is recommended when calling this function:

- It is recommended to indicate that processing is in progress on the SDK side, e.g. by displaying a loading message.

- Clearly display error information that is important for the user, e.g. wrong PIN

Only Comply to "He" method is supported as an KYC verification method.

※Comply to "He" method + (My Number); 'Comply to "To" method (via SDK); etc. are not supported.

Only 'driver's licence,' 'National Identity Card,' 'Residence card,' and 'Special permanent resident certificate' are supported as ID document types.

※National Identity Card (+ My Number)' is not supported.

Please contact our sales team for more information

Notes

- For the half-modal part, the following 4 types of half-modals are displayed during the process of end-user operation



※UI details differ depending on the iOS version

- These half-modals are forced to be displayed according to standard iOS specifications and are automatically closed depending on the operation status
 - ※The Android version of the LIQUID eKYC SDK is implemented in accordance with the above behaviour on iOS
 - Therefore, error screens, etc. prepared by the client must be displayed separately from the half-modal screen
 - ※The error screen prepared by the client is displayed after the half-modal display
 - In Android, the expression of the half-modal part can be changed arbitrarily
 - ※The expression of the half-modal part of the iOS cannot be changed
- Overwrite the layout with the same resource name to replace the half-modal screen design described above

- ✧ To change half-modal ①
Prepare layout/liquid_layout_chip_reading_bottom_sheet_content_for_ready.xml and implement the design of your choice
- ✧ To change half-modal ②
Prepare layout/liquid_layout_chip_reading_bottom_sheet_content_for_reading.xml and implement the design of your choice
- ✧ To change half-modal ③
Prepare layout/liquid_layout_chip_reading_bottom_sheet_content_for_success.xml and implement the design of your choice
- ✧ To change half-modal ④
Prepare layout/liquid_layout_chip_reading_bottom_sheet_content_for_failure.xml and implement the design of your choice

Please cooperate with displaying the statement "Cybertrust's iTrust(R) is used to determine the authenticity of ID cards" on the client's screen

Please contact our sales team for more information

Precautions - When using "Residence card" or "Special Permanent Resident Certificate"

- When using the Comply to "He" method with "Residence card" or "Special Permanent Resident Certificate", an image of the back side of the card is required.
- The image of the back side of the card must be captured on the client's side and submitted via the "Document IC Card Supplementary Info Transmission Function (without screen)"
 - Specifications for the image of the back side of the card :
 - ✧ Resolution : Up to 1920 by 1920 pixels

For details on interfaces other than those mentioned above, please refer to the attached "Interface Specifications for LIQUID eKYC Applicant SDK Version".

Document IC Card Supplementary Info Transmission Function (without screen)
"verifyIdChipSupplementaryNonInteractive()"

Subsequent functions when "Document IC Card Supplementary Info Transmission Function (without screen)" is used with a Residence card or Special permanent resident certificate.

The image of the back side of the card taken by the client is transferred, and the remaining processing related to the Comply to "He" method is executed.

For details on interfaces other than those mentioned above, please refer to the attached “Interface Specifications for LIQUID eKYC Applicant SDK Version”.

JPKI Function “identifyIdChip()”

The Public Certification Service for Individuals using Individual Number Card.

Reads the information on the chip embedded in the Individual Number Card to do Public Certification. The process returns the name, date of birth, address, and sex of the card’s owner.

Precautions

- “JPKI evidence” is kept within Liquid, but if you wish to hold it within your organization’s system as well, save the “JPKI evidence” returned by this function or the “JPKI (Public Certification)” returned by “LIQUID_eKYC_022_GetICCardInfo_API”. However, when deleting this data, you must make sure there is a backup copy within your organization.
- Individual Number Cards with special date of birth expressions cannot not be read.
e.g. 2020/A1/A3 (at the end of spring in 2020), 0000/01/00 (on an unknown year, January, on an unknown date)
- Individual Number Cards for testing, provided by the Japan Agency for Local Authority Information Systems, may only be used on weekdays from 10 AM to 6 PM.

For iOS, “AID information” needs to be specified on Xcode.

“AID information” is a fixed character string on Driver’s Licenses, Individual Number Cards and Residence Cards/ Special Permanent Resident Certificate.

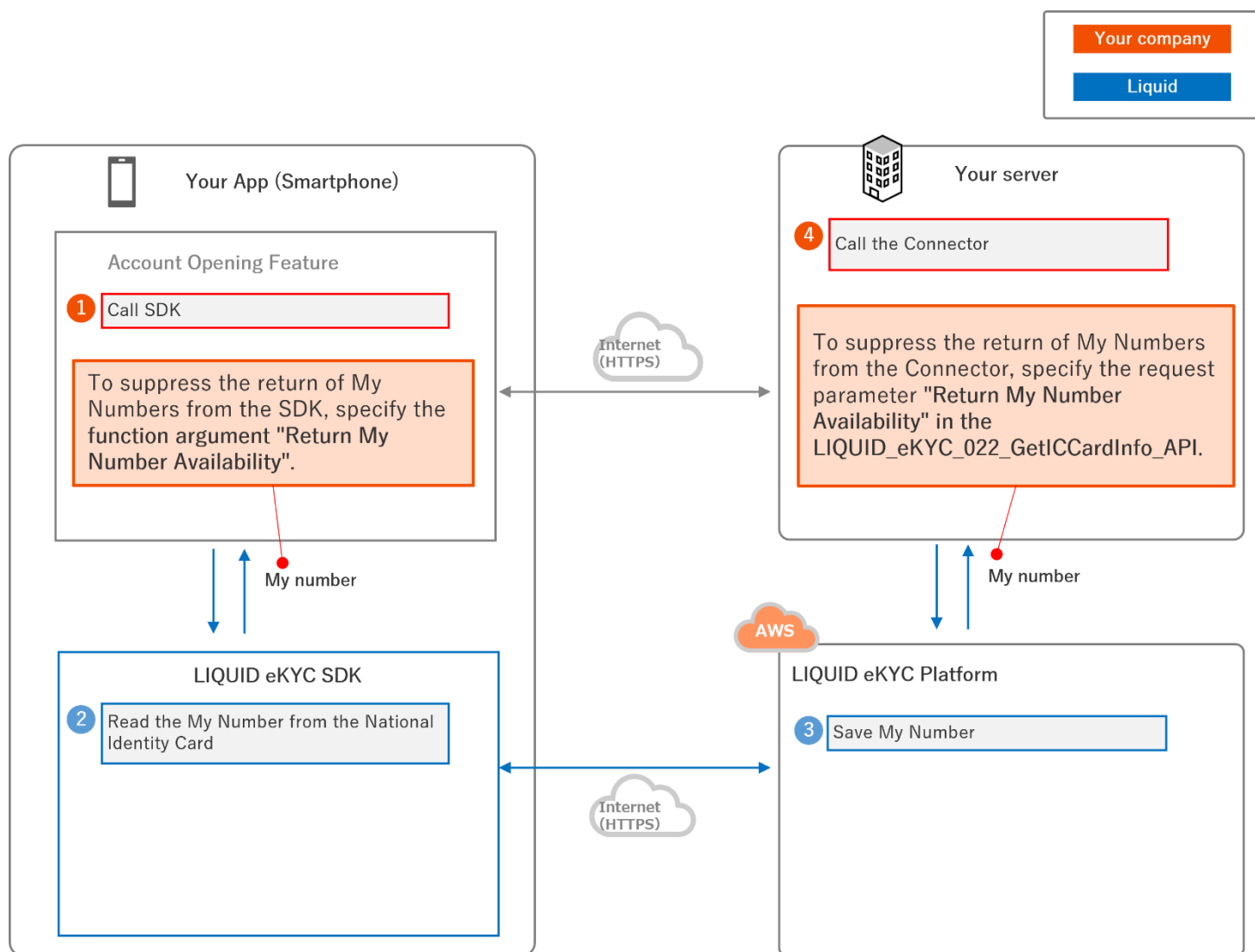
For Individual Number Cards however, a NDA must be signed for disclosure of “AID information”.

**The NDA process will take around two to four weeks to complete.*

Please contact our sales team for more information.

Control of My Number Acquisition for National Identity Cards

The return control method to the client's app or core system is as follows:



For details of the interface, please refer to the "LIQUID eKYC Applicant SDK Interface Specifications".

JPKI function (without screen) "identifyIdChipNonInteractive()"

Functionality of 'JPKI Function' remains unchanged, with the function providing screens only for the chip reading portion.

The chip reading display (henceforth referred to as "half-modal") is superimposed on the screen of the client's app, and communication takes place asynchronously.



Therefore, the following is recommended when calling this function:

- It is recommended to indicate that processing is in progress on the SDK side, e.g. by displaying a loading message.
- Clearly display error information that is important for the user, e.g. wrong PIN

Notes

- For the half-modal part, the following 4 types of half-modals are displayed during the process of end-user operation



※UI details differ depending on the iOS version

- These half-modals are forced to be displayed according to standard iOS specifications and are automatically closed depending on the operation status
※The Android version of the LIQUID eKYC SDK is implemented in accordance with the above behaviour on iOS
- Therefore, error screens, etc. prepared by the client must be displayed separately from the half-modal screen
※The error screen prepared by the client is displayed after the half-modal display
- In Android, the expression of the half-modal part can be changed arbitrarily
※The expression of the half-modal part of the iOS cannot be changed
 - Overwrite the layout with the same resource name to replace the half-modal screen design described above
 - ✧ To change half-modal ①
Prepare layout/liquid_layout_chip_reading_bottom_sheet_content_for_ready.xml and implement the design of your choice
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- ✧ To change half-modal ③
Prepare layout/liquid_layout_chip_reading_bottom_sheet_content_for_success.xml and implement the design of your choice
- ✧ To change half-modal ④
Prepare layout/liquid_layout_chip_reading_bottom_sheet_content_for_failure.xml and implement the design of your choice

Please cooperate with displaying the statement "Cybertrust's iTrust(R) is used to determine the authenticity of ID cards" on the client's screen

Please contact our sales team for more information

For details on interfaces other than those noted above, please refer to the attached "LIQUID eKYC Applicant SDK Interface Specifications".

Japanese Public Key Infrastructure (smartphone JPKI) function "identifyIdMyna()"

Japanese Public Key Infrastructure process using the Myna Portal application.

In addition to reading pre-registered (or physical my number card) data and performing Japanese Public Key Infrastructure with the Myna Portal application, four basic pieces of information (name, birthday, address, and sex) can be obtained as a result of the process.

Note

- The trail of Public Key Infrastructure is stored by Liquid, but if you wish to store it on your side as well, please store the "Trail of public personal authentication" returned by our function, or the "Public Key Infrastructure application" of the eKYC Connector's "GetICCardInfo_API". However, if you wish to delete the data, you are required to store it on your side.
- Cards with special birth dates are not available.
 - Example: 2020/A1/A3 (late spring of 2020), 0000/01/00 (unknown date in January of an unknown year), etc.
- The hours of availability of the test National Identity Card that can be borrowed from the Organization of Local Public Entities Information System are weekdays from 10:00 to 18:00.

Various actions such as application to the Digital Agency are required to use the Public Key Infrastructure function (Smartphone JPKI) function.

Please contact our sales staff for details.

For Android only. For inquiries regarding the implementation timeline for iOS, please contact our sales representative.

For interface details, please refer to the attached "Interface Specifications for LIQUID eKYC Applicant SDK Version".

Card-Substitute Electromagnetic Record function (AppleWallet) "verifyIdWallet()"

Retrieval of National Identity Card information registered in Apple Wallet and Signature Verification Processing.

The processing results allow retrieval of the "basic four" data (name, date of birth, address, and gender) as well as a facial photo.

For interface details, please refer to the attached "LIQUID eKYC Applicant SDK Interface Specifications".

Facial Recognition Function "verifyFace()"

Face scan and verification.

There are two methods of determination.

1. Active Detection : Requires additional operations by the user and flash photography)
2. Passive Detection : No additional operations by users themselves required(facial muscle * paid option

verification method	Face photo (Front)	Facial muscle photo *Temporary mouth opening action	Flash photo *Holding the device closer to the face Operation
Active detection	Yes	Yes	Yes
Passive detection	Yes	No	No

A screen for reviewing the photos taken (users can review the photos themselves and decide whether to re-take them) is also provided.

*The use of this screen is optional but recommended.

For details of the interface, please refer to the "LIQUID eKYC Applicant SDK Interface Specifications".

OCR Result Retrieval Function "getOcrResults()"

Reads in text from documents.

This activates data transmission within the function. However, the SDK does not require this process to succeed in order to go on to other functions. Therefore, extra UI implementation and error handling is up to each client's discretion.

For details of the interface, please refer to the "LIQUID eKYC Applicant SDK Interface Specifications".

Additional Document Photo function "captureOptionalDocument()"

Process for photographing any document

For interface details, please refer to the attached "LIQUID eKYC Applicant SDK Interface Specifications".

For a detailed explanation including screen flow, please refer to the Appendix '(SDK) Additional Document Shooting Options Description'.

Activate Function "activate()"

Confirms data on the eKYC Platform.

Please note that when calling this function, clients will need to prepare the following. This is necessary because the function activates communication between systems.

- A UI that notifies the user about the data transmission (e.g. a loading screen).
- A rerun of the function when the process fails (e.g. connection failures). For example, prompting the user to rerun through a dialog.

For details of the interface, please refer to the "LIQUID eKYC Applicant SDK Interface Specifications".

IC card reading availability function "checkIdChipAvailability()"

Process to check if the terminal is NFC compatible.

In the case of devices that do not support NFC, meaning they cannot read IC cards, the following functions are recommended for use, excluding the functions listed below

- "Document IC Card Function"
- "Document IC card function (without screen)"
- "JPKI function"
- "JPKI function (without screen)"

For interface details, please refer to the attached "LIQUID eKYC Applicant SDK Interface Specifications".

Selecting version function “getVersion()”

The SDK version can be selected.

For interface details, please refer to the attached "LIQUID eKYC Applicant SDK version Interface Specifications".

Please use for inquiries, etc.

Display language change function “changeDisplayLanguage()”

The display language in the normal circumstance depends on the terminal settings, but this function allows the display language to be specified explicitly. The following display languages can be specified.

- Japanese
- English
- Vietnamese
- Indonesian
- Simplified Chinese
- Portuguese
- Korean
- Nepali
- Burmese
- Khmer
- Mongolian
- Malay
- Thai

For interface details, please refer to the attached "LIQUID eKYC Applicant SDK Interface Specifications".

Please execute before the function with screen display, as it applies to all functions with screen display once it's executed.

Note that the display language cannot be specified in this function for text that is not under the control of the LIQUID eKYC SDK.

Precautions for Android

In some cases, the SDK running within the client's app may not display the screen in the language specified by this function. This is due to the removal of language resources when downloading the app from Google Play.

To avoid this issue, it is recommended to configure the app to always install all language resources.

◇ Incident Example

Although “Portuguese” was specified in this function, each SDK function displays content in English.

◇ Cause

When downloading apps from Google Play, if there are languages that are not configured on the device, the in-app language resources for those languages will be removed.

◇ Countermeasure Example

Configure the app to always install all language resources :

```
android {  
    bundle {  
        language {  
            enableSplit = false  
        }  
    }  
}
```

Design setting function “customizeDesign()”

Button color can be set.

※ The buttons at the bottom of the Terms of Use function are not affected by the settings in the Design Setting function. If you wish to change the colors in the Terms of Use screen, please set it through the argument of the Terms of Use function.

For interface details, please refer to the attached "LIQUID eKYC Applicant SDK Interface Specifications".

Please execute before any function with screen display (one time)

Header Font Color Settings

A setting that changes the font color of the text outside the frame.

* Note that this is not a function.



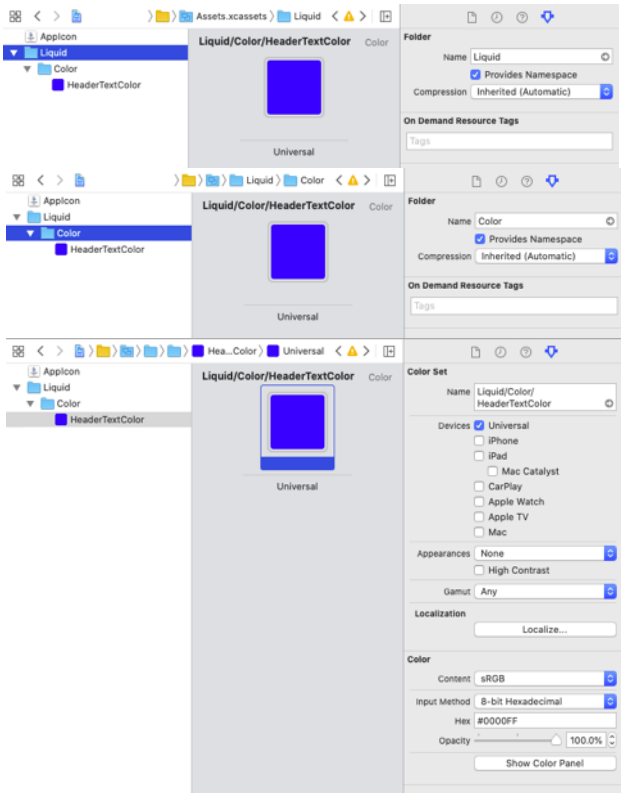
Values to be given to the SDK

Modify the settings within the SDK's project.

Content	Type	Data Format	Max Characters	Required	Details
New font color	Shown below			△	* The initial value is #FFFFFF (white)

- iOS
 - Mainly for camera screen
 - ✧ Define the color set within Assets as Liquid/Color/HeaderTextColor

Example:



- Mainly for screens other than camera (e.g. chip reading in Comply to “He”)
 - ✧ Define the color set within Assets as Liquid/Color/HeaderTextColorForWhiteBackground

- Android
 - Mainly for camera screen
 - ✧ Define the color resource using the name R.color.liquid_header_text_color
 - ✧ Example: Prepare values/color.xml and define as below.


```
<color name="liquid_header_text_color">#0000FF</color>
```
 - Mainly for screens other than camera (e.g. chip reading in Comply to “He”)
 - ✧ Define the color resource using the name R.color.liquid_header_text_color_for_white_background

Return values from the SDK

None

Common Return Values: Process Results

All the functions return values in the same format as long as they are “process results”.

To be more specific, the functions return the “Names (enum)” and “Error code” below.

- “Names (enum)”: Error types for handling. We recommend using these values and not the error codes for error handling.
- “Error code” : Used when communicating with Liquid.

Please refer to the "LIQUID eKYC Applicant SDK Interface Specifications Sheet: Various process results ".

Common Return Values: Additional Information (Server Maintenance)

When the process results are “Server maintenance (maintenance)”, all the functions return additional information.

Please refer to the "LIQUID eKYC Applicant SDK Interface Specifications Sheet: Server Maintenance".

SDK Standalone Function 【Liquid specified. For special purpose operators】

When using the Face Verification function in a client's app, the following flow must always be followed:

- Execute the Initialization Function (for standalone function) and ensure it completes successfully.
- Execute the Face Verification function and ensure it completes successfully.
- Execute the Activate function and ensure it completes successfully.

For information on how to obtain and integrate the SDK, please refer to the separately provided README.

For specific function specifications, please refer to the separately provided API Documents.

Initialization Function (for standalone function) “startVerifyStandalone()”

Preparatory processing to enable the Face Verification function

Note that because this function involves network communication, the following two points must be observed when calling it:

1. Prepare a UI to inform the user that communication is in progress (e.g. Display a loading indicator)
2. When processing fails (e.g. Network disconnection) , Execute this process again (e.g. Display a dialog and prompt the user to execute again)

For interface details, please refer to the attached "LIQUID eKYC Applicant SDK Interface Specifications".

Face Verification function “collateFace()”

Verification processing of two facial images.

Since no capture functionality is included, prepare the facial image on the document surface and the selfie facial image in advance before calling this function.

Note that because this function involves network communication, the following two points must be observed when calling it:

1. Prepare a UI to inform the user that communication is in progress (e.g. Display a loading indicator)
2. When processing fails (e.g. Network disconnection) , Execute this process again (e.g. Display a dialog and prompt the user to execute again)

For interface details, please refer to the attached "LIQUID eKYC Applicant SDK Interface Specifications".

eKYC Connector

For security purposes, sensitive data can only be transmitted through system linkage.

The details are provided in a separate document for eKYC Connector